

GRA-Core: Hierarchical Stability for Subject-Oriented AI Swarms

Orchestra Encourages Local Foam, Nullifies Global Foam

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Abstract

We introduce GRA-Core, a universal hierarchical stability framework for orchestrating AI agents with preserved individuality. The key principle: *The orchestra encourages the foam of AI agents, but nullifies the foam in the orchestra.* Local foam (agent’s internal contradictions, quirks) is allowed and even rewarded, while global foam (collective inconsistency, goal violation) is strictly nullified through gradient-based adaptation. We formalize the separation, provide convergence conditions, and demonstrate a working implementation with backend API and interactive dashboard.

1 Introduction

Modern AI swarms face a dilemma: enforcing global coherence often kills agent diversity. We propose a hierarchical stability functional where each agent retains a non-zero “local foam” $\Phi_i^{\text{local}} > 0$ (their character, biases, creative noise), but the orchestra as a whole must satisfy $\Phi_{\text{global}} = 0$ (perfect alignment with the global goal).

2 Formal Framework

Let x be the global state of the swarm, and a_i, s_i the state and personality parameters of agent i . Define the total functional:

$$J(x, s_{1..N}) = \Phi_{\text{global}}(x) + \sum_{i=1}^N \left(\alpha_i \Phi_i^{\text{local}}(a_i, s_i) + \beta_i \Phi_i^{\text{align}}(a_i, s_i, x) + \gamma_i \Phi_i^{\text{reg}}(a_i, s_i) \right) \quad (1)$$

- $\Phi_{\text{global}}(x)$ – global foam (must be nullified).
- Φ_i^{local} – agent’s internal foam (allowed, bounded by c_i).
- Φ_i^{align} – alignment violation with the orchestra.
- R – regularizers.

The GRA-Core update rule:

$$(x^{t+1}, s_{1..N}^{t+1}) = (x^t, s_{1..N}^t) - \eta \nabla J(x^t, s_{1..N}^t), \quad (2)$$

with the constraint that gradient components corresponding to Φ_i^{local} are damped to preserve $0 < \Phi_i^{\text{local}} \leq c_i$.

3 Stability Conditions

A swarm is called *stable* iff:

$$\Phi_{\text{global}}(x^*) = 0, \quad (3)$$

$$\Delta \Phi_{\text{global}}(x^*) = 0, \quad (4)$$

$$\Delta^2 \Phi_{\text{global}}(x^*) > 0, \quad (5)$$

$$0 < \Phi_i^{\text{local}}(a_i^*, s_i^*) \leq c_i \quad \forall i. \quad (6)$$

The first three conditions ensure the orchestra is at a local minimum of global foam; the last guarantees each agent retains its personality.

4 Implementation

- (1) We provide a reference implementation with:

- FastAPI backend – maintains agents, computes foams, performs GRA steps.
- Frontend dashboard – visualises local/global foam, allows connect/disconnect of agents.
- Agents are stored with their character vectors s_i and can be moved between swarms.

5 Experimental Demonstration

Using the demo agents (impressionist, paranoid, disciplined, rebel), we start with non-zero global foam. After each GRA step, Φ_{global} decreases while each Φ_i^{local} fluctuates but stays above zero. The system reaches $\Phi_{\text{global}} < 10^{-2}$ within 5–10 steps, and the stability criteria are satisfied.

6 Conclusion

GRA-Core enables truly living AI orchestras: agents keep their quirks and creativity, yet the ensemble remains perfectly coherent. The framework supports detachment, reattachment, and composition of heterogeneous swarms.

References

- [1] GRA-Core Unified Hierarchical Stability Library, <https://github.com/qqewq/GRA-Core-new-Unified-Hierarchical-Stability-Library>